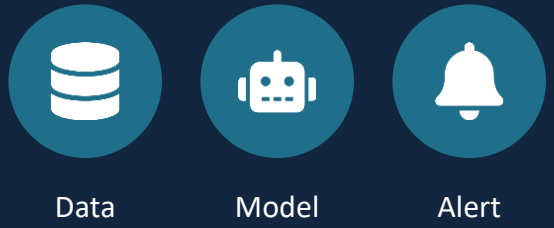
Bank Transaction Fraud Detection Using Python

A Machine Learning mini project to classify transactions as genuine or fraudulent





Introduction



Digital banking is growing

More online payments also mean more fraud risk.



Old methods fall short

Fixed rules and manual checks miss new fraud patterns.



Machine Learning helps

It finds unusual patterns in large transaction data.

Project Aim

Build a Python system that predicts if a transaction is genuine or potentially fraudulent.

Abstract

Uses Pandas, NumPy, Matplotlib and Scikit-learn with Logistic Regression or Random Forest.

Objectives & Requirements

- Understand transaction fraud detection
- Analyze banking transaction datasets
- Find suspicious transaction patterns
- Apply ML algorithms to classify fraud
- Label transactions genuine or fraudulent
- Evaluate the trained model
- Improve digital banking security awareness

Software

- Python
- VS Code / Jupyter / Colab
- Labeled transaction dataset

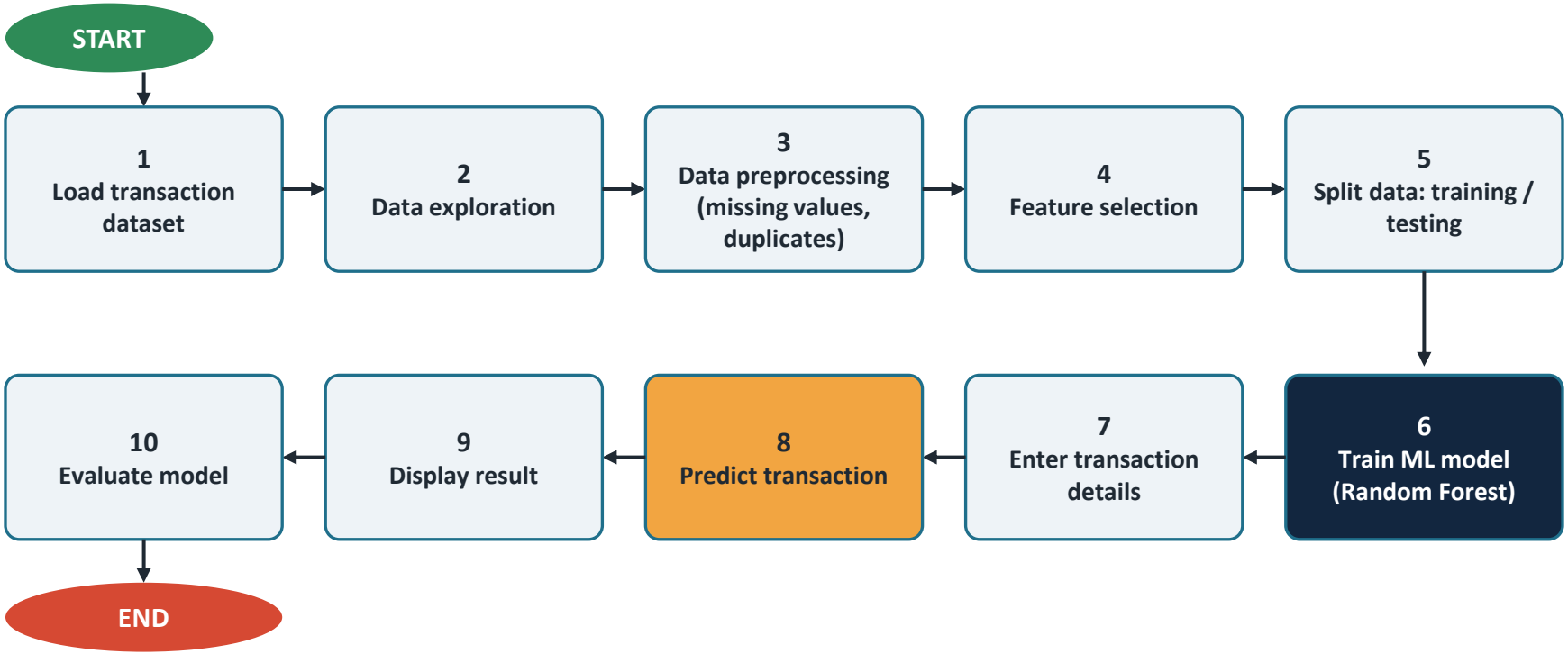
Hardware

- Computer or laptop
- Minimum 4 GB RAM
- Internet connection

Algorithm

1. Import Python libraries and load the dataset
2. Explore data; remove duplicates and fix missing values
3. Select relevant transaction features
4. Split data into training and testing sets
5. Train a classification model
6. Give transaction details to the model
7. Predict genuine or fraudulent and show the result
8. Evaluate model performance, then stop

Flowchart


START
1
Load transaction dataset
2
Data exploration
3
Data preprocessing (missing values, duplicates)
4
Feature selection
5
Split data: training / testing
6
Train ML model (Random Forest)
7
Enter transaction details
8
Predict transaction
9
Display result
10
Evaluate model
END

Methodology

Input module
ID, amount, type, time of transaction

Processing module
Clean data, encode, select features, split

Decision module
Genuine or potentially fraudulent

Looping module
Repeats for every transaction record

Output module
Status, amount and model metrics

Python libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Result: Sample Output

```
===== BANK FRAUD DETECTION =====
1. Add Transaction
2. View History
3. View Summary
4. Exit
Enter choice: 1
Transaction amount: 90000
Failed attempts: 1000
New device? (yes/no): yes
Result: FRAUD - SUSPICIOUS
- High transaction amount
- Multiple failed attempts
- New device detected
```

Normal transaction
Amount 5000 on a known device: NORMAL.

Suspicious transaction
Amount 90000 on a new device: FRAUD.

Summary view
Total 2: 1 normal, 1 suspicious.

Conclusion & Future Work

Conclusion

- Machine Learning can find possibly fraudulent banking transactions.
- Python libraries make data work and modeling easier.
- It is a prototype, not a complete real-world security solution.

Future Work

- Real-time detection
- Deep learning
- Web app (Flask / Streamlit)
- Database integration
- Better feature analysis
- Model improvement
- Alert system
- Explainable AI